Love In the Cloud

Installation | Human Computer Interaction | App Design



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Love In The Cloud provides a reciprocal, intuitive, and cozy hugging experience that meets people's psychological needs, bringing people emotional comfort and support via Fluffy, an installation. The hugging system allows people to express their affection for each other and strengthen connections, despite their geographical distances.

Inspiration from My Life

- Hugs help express affection and ease the tension after fighting with my intimate partner.
- Hugs help express concern and care for friends
- Hugging is empowering and encouraging when I feel sad.
- Free hugs are available and needed everywhere.

Research

Psychologists find the importance of maternal touch in infant development

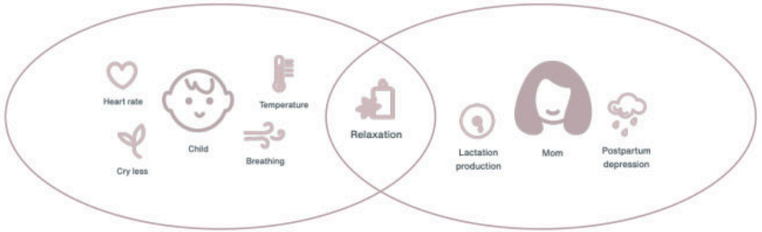
Case 1: Experiments shows that closeness brings affection and security to infants.

Psychologist, Harry Harlow, have revealed the importance of maternal contact on child development. In the experiment, newborns were removed from their biological mothers immediately after birth. They were given two inanimate surrogate mothers: one was wired but offered milk, the other one was made with cloth but offered no food (Harlow & Zimmermann, 1958).



Harlow's experiments also find that, if reared in isolation after birth, infant monkeys showed bizarre behavior such as staring blankly and shaking compulsively. When placed back to peers, the monkeys did not know how to socialize with the other monkeys. Some of them were scared and became aggressive towards the other monkeys. The other monkeys bullied them, which made them end up in tearing hair out, scratching, and self-mutilation. Many of them stayed isolated, while the others died due to refusal to eat. (Harlow & Dodsworth et al., 1965).

Case 2: Skin-to-skin time has positive impact on babies as well as mothers (Greicius, 2013).



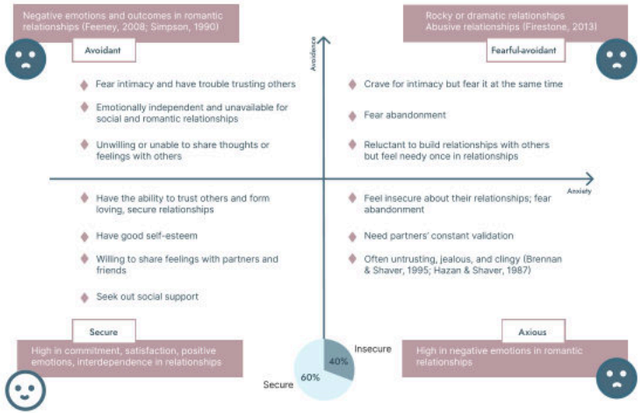
Case 3: Touch impacts babies' attachment style, which has a life-long impact on various aspects of their adults' life.

Attachment style is how a person relates to others in relationships. It is formed through interactions between children and their primary caregivers.

The secure attachment is formed when babies' primary caregivers are responsive to their needs, both physically and emotionally (Bowlby, 1988). As a foundation to secure attachmen, touch connects parents with infants, strengthening their bonds (Heller, 1997).

In the absence of touch, children developes maladaptive behaviors such as hostile aggreesion, excessive crying, and difficulties eating food, forming insecure attachment styles: anxious, avoidant, and fearful-avoidant(Davis, Rovers & Petrella, 2017).

Four types of attachment style



Hugs, as a form of affective touch, benefit people's wellbeing

1. Hugs help human communicating emotions and maintaining relationships

Touch is a fundamental language that helps people connect, trust and collaborate (Mccluskey, 2020). Negative impact of interpersonal conflict such as disagreements and argument can be smoothed through hugs (Murphy, Janicki-Deverts & Cohen, 2018).

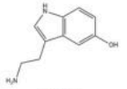
2. Hugs provide psychological and physical benefits

Hugs positively affect people's health by influencing the 'happy hormones', including serotonin, depamone, and cortisol.

Psychological benefits

- stabilize mood
- reduce depression
- regulate anxiety

Hormones



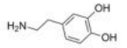
Serotonin

Physical benefits

- regulate digestion
- heal wounds
- maintain bone health
- lessen sleep disruptions

Dopamine contributes to feelings of (Pietrangelo, 2019)

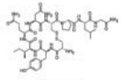
- alertness
- focus
- motivation
- happiness



Dopamine

- stimulate nausea
- decrease pain
- coordinate movement (Mandal, 2019)

- relaxation
- trust
- overall psychological stability
- bonding



Oxytocin

- help body adapt to different emotional and social situations

Psychologically, hugs lessen feelings of isolation and loneliness, and activate parts of brain that help people empathize. Physcially, hugs calm the nervous system, reduce the stress level, lower blood pressure and heart rate, and improve healing (Scaccia, 2020). Hugs even strengthen the immune system, making people less likely to catch the common cold (Cohen, Janicki-Deverts, Turner & Doyle, 2015).

♥ However, people's needs cannot be met in various situations

Long-distance relationship

Long-distance relationship leaves a physical gap between couples, breeding loneliness and a lack of intimacy and emotional support, which can be fulfilled through touch (Betchen, 2015).

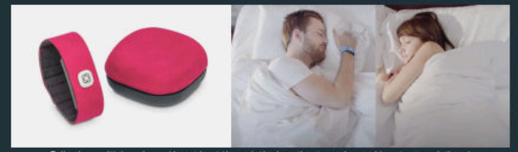
Bumpy relationships can be even harder to maintain because it's impossible to comfort each other through touch (Tower, 2016), such as a hug, a touch on the cheek, or a rub of a shoulder.

Distances create more misunderstanding due to a lack of body language in the communication, which makes the energies emitted by emotions unperceived. At the same time, it is impossible to adjust perceptions or correct misunderstanding through touch. At the same time, people are trying to fill the gap, making people feel physically closer through devices connecting senses (Neustaedter, 2020).



A person can feel the other's touch through bent fingers

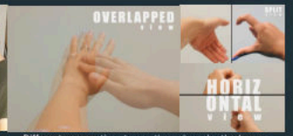
Show miss through the wristband's vibration



Fall asleep with loved ones' heart beat through the heartbeat speaker and heart rate wristband



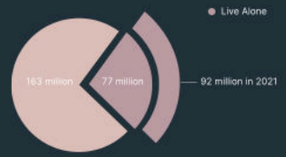
See through the other's eyes



Different perspectives to see through each other's eyes

Singlehood

Singlehood becomes the tendency worldwide. In major cities like Munich, Frankfurt, and Paris, more than 50% of families are single-person households. In the U.S., the number of singles rises from 22% in 1950 to more than 50% nowadays. The percentage of singles is predicted to rise 20% worldwide in 2030 (Kislev, 2019).



China Statistical Yearbook on China's Singles in 2019

People face the challenges of loneliness and the needs of company. Loneliness makes 67.8% of singles want to stop living alone as soon as possible, stated by the Report of Young Adults Who Live Alone in 2020.

Lack of physical touch for a prolonged amount of time can cause touch starvation, which makes people feel overwhelmingly lonely or deprived of affection (Sharkey & Lamoreux, 2021).

People may also experience
- stress
- anxiety
- feelings of depression
- low satisfaction
- difficulty sleeping
- fatigue

'Vegetarian hook up' becomes notable in China as a solution to fill the needs of touch and company. 'Vegetarian hook up' is hook up for sleep, not for sex, so that people can meet their needs of touch through cuddling. The concept first appears in Japan around 2013. These people are called sleeping partners ソフレ.

Infectious disease

Staying home and away from others help prevent spread of COVID-19. However, **Social distancing causes a lack of physical contact that grows loneliness and social isolation, which leads to touch starvation (Pierce, 2020).**



Valentina Ogaryan
Clinical psychologist

"Many patients are expressing **profound sadness** about this and saying **how isolated they feel**. Those who are older and don't live with a lot of family members say they even **miss the more minor and nuanced encounters** that used to bring a touch here and there, like a handshake, smile, or pat on the hand or back from a neighbor or someone at the grocery store."

"It is normal to struggle with the loss of touch. Touch is a legitimate physical and emotional need. It's part of the human experience, and **losing that and not knowing when you will get it back is hard**."



Devita Streva
Psychotherapist

People attempts to find solutions around the world.



'Hug room' in Italy let nursing home residents touch their family after months of isolation (Pitrelli, & Harlan, 2020).



'Hug curtain' allows an 85-year-old Brazilian woman getting her first embrace after five months' isolation (Seymour, 2021).



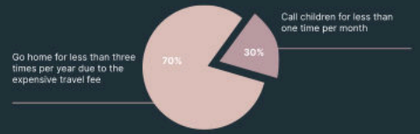
People in Israel combat with the COVID blues by hugging trees (Heller, 2020).



Plastic hug window in Italy allows COVID-19 patients hug their loved ones (Plastic hug wall, 2021).

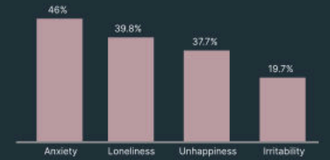
Stay-at-home children

In China, more than six million children are raised by grandparents or family friends in rural regions while their parents work in the cities. Children face developmental and emotional challenges growing up with limited interaction with their parents.



Parents' frequency to take care of their children

China Youth and Children Research Center finds that stay-at-home children face more negative emotions than the children raised by their parents.



Negative emotions of stay-at-home children

However, developmental challenges can be eased through communication. During the talk, children can receive parents' emotional support; they feel happier after the conversation.

With the development of society, the population becomes more mobile than ever before. Hugging someone we love becomes more difficult. Also, people's opinion about marriage has changed, with more people choose to remain single, which make them face the challenges of loneliness and touch starvation. At the same time, diseases keep people apart, preventing patients getting the emotional support they need. **Therefore, filling the need becomes more urgent.**

People make effort to provide hugs and touch

As many people have noticed the need, effort are made to allow people touch or hug each other over distances. With the development of technology, researchers strive to make the haptic interfaces thinner, lighter, and more comfortable, enjoyable to wear, as if the devices are people's second skins. In the market, interactive products allow people to send hugs to the other, and noninteractive products provide a channel for a cozy, quiet moment of hug.

1. Non-interactive products provide a channel for hugging; However, they are not as fulfilling as interactive hugs

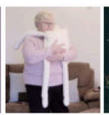


Products and installations provide objects for people to hug. Researchers find that even hugging a simple, inanimate object reduces stress (Sumioka, Nakae, Kanai & Ishiguro, 2013). Touching an inanimate objects that mimic human touch also helps soothe existential fears in people with low self-esteem (Koole, 2013). However, these products are not interactive. Since people hug to express affection, full reciprocal hugs can be more fulfilling and instinctive.

2. Interactive products allow people to send hugs to their love ones remotely, but it is one-way communication



Hug toy for children


Hug toy for elders


Hug clothes


Facebook's inflatable hug clothes


Hug installation

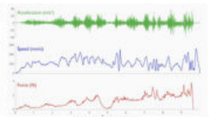
By clicking an icon on apps or rolling the handle beside the installation, one person can send a hug to another through devices, either it is a stuffed toy or a piece of clothes.

In most cases, it is a one-way communication, as people receiving hugs can hardly hug back, let alone people hug each other in real-time.

3. Haptic technology allows people touch over distances, but the touch area is limited, let alone hugs


Haptic skin with air inflatable materials


Transfer touch through screen


Data recorded


Haptic skin helps prosthesis users regain sense of touch


Haptic skin vibrate to mimic touch


Pen replays the data when in use

Haptic technology allows people to touch each other over distances by applying forces, vibrations, or motions to the users. Such data is collected through tactile sensors that measure forces exerted by the user on the interface. Data is a precise way to replay the touch on the other end, and it is widely used in fields such as communications, entertainment and medicine. However, the touch area is limited to mimic a hug (Sardin, Serpelloni & Pasqui, 2014).

4. Robots provide interactive, face-to-face hugging experience but limited emotional support


HuggieBot 1.0


HuggieBot 2.0

Researchers have created a life-sized humanoid hugging robot that aims to provide emotional support to people, wherever or whenever they need it. With the 1.0 version made in 2018, the 2.0 version has just come out in February 2021, with updated shape, sensors, and softer materials to give people a more secure and adaptive hugging experience. **It is reciprocal, but it lacks psychological comfort.**

Ideation

Due to geographical constraints, people cannot receive emotional support from friends and family in time; they also lack a channel to express their affection precisely. The existing products can hardly provide reciprocal experience or psychological support.

I want to provide people with emotional comfort and support immediately when they are in need through reciprocal, intuitive, and cozy hugs; the hugging system will allow people to express their care and love precisely and connect, anywhere, anytime, over geographical constraints no matter if they are couples, family members, or friends.

Design matrix



Emotional Support: cozy, friendly, comfortable, soft, and safe feelings.

Timeliness	▷ People can immediately receive support from friends and family when they are in need. They can hug their loved ones to comfort them as
Transfer emotions precisely	▷ Data will be recorded and transmitted precisely to convey emotions, love, and care.
Reciprocal Experience	▷ People can interact with each other through the devide ▷ Have different modes considering different scenarios

Technology adoption

To hug a person, arms need to bend, holding the person without squeezing them. Therefore, I have found three technology solutions: robotic arm, air inflatable materials, and soft robotics. I rate the three solutions based on the design matrix and economic constraints, including the total cost and the difficulty of realization. Criteria's weight is based on how much it impacts the desired user experience.

The installation can be either placed in the public space or at home. If served for public service, Fluffy will become a public infrastructure storing in the booth to enhance public health. Thus, the Government will fund project construction.

Based on the result, I decide to use air inflatable materials.

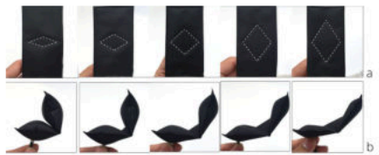
			Desirable 1 → 5		
Criteria		Weight	Robotic arm	Air inflatable	Soft robotics
Emotional Support 60%	Timeliness	25%	5	5	5
	Preciseness	15%	3	4	5
	Reciprocal	20%	5	3	5
			2.7	2.45	3
Economic 40%	Cost	20%	1	5	1
	Tech Difficulty	20%	1	5	1
			0.4	2	0.4
Total			3.1	4.45	3.4

Structure test

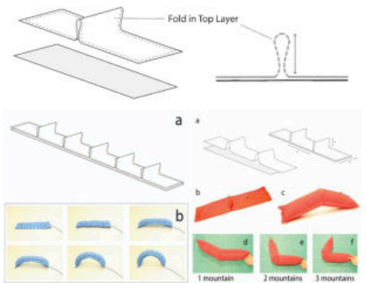
Structure Inspiration



Therms-Up! DIY Inflatables by MIT Media Lab



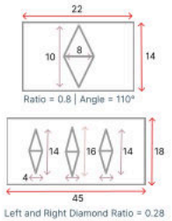
aeroMorph by MIT Media Lab

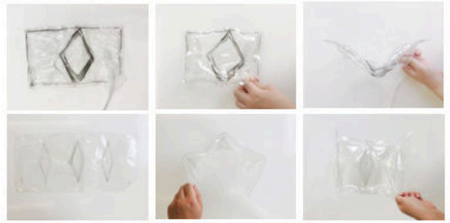


Printflatables by MIT Media Lab

Structure Inspiration

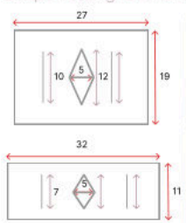
Shape Testing: diamond

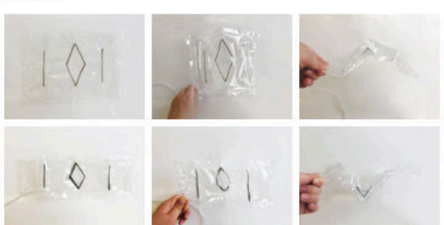




Inspired by MIT's Printflatables and Therms-Up!, I have identified diamond as a solution to make the arms bend. Therefore, I have made prototypes to test the structure. As found in aeroMorph of MIT, the ratio of the diamond's width and height hinge determines the inflatable's bending angle at full inflation. I made the ratio small to make the inflatable bends with greater angle. I find that with ratio between 0.25 and 0.28, the bending angle is 60°. The ratio of 0.8 makes the inflatable bends at 110°. Thus, I can make the arms bend by combining diamonds to determine the bending angle.

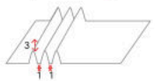
Shape Testing: diamond + line

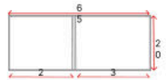




As shown by (a) in Therms-Up!, I have also tested diamond and lines. Although lines can make the inflatable bend, the lines do not provide directional support for angles. The bending direction can easily be changed due to the outside force.

Shape Testing: mountains







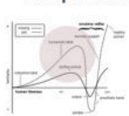
As the prototypes show, it does not bend at ideal angle. The height of the mountains need to be bigger for the inflatable to bend at a bigger angle; the distance of the mountains can be bigger so that each mountains could provide bigger angle at full inflation. It is a feasible solution, with more testing needed.

Design

Color, Materials, Finish

Soft Touch	▷ As Harlow's experiment, people prefer soft things for comfort ▷ Soft material mimics the touch of human
Round Shape	▷ Round shape offers senses of softness, friendliness, safty, and security because it does not have angles (Design, 2020)
Human Size	▷ Nonportable product for home use ▷ Mimics the hugs between people ▷ Mimics bear hugs to offer cozy feelings as they usually shared between family members, close friends or lovers (Quinn, 2017)

Shape ideation



The Uncanny Valley Theory



Soft robotics by CMU



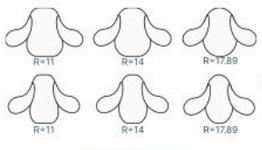
Humanoide



Yellow Sculpture

To avoid unease or revulsion caused by human-like appearance (Cherry, 2020), I brainstorm based on stuffed toy and humanroid robot to ensure people's likeness.

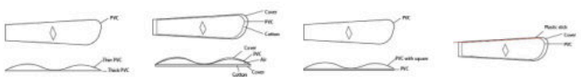
To find the most lovable shape, I have tested different angles and the size of the body.



Shapes brainstorming



Finalized Shape



Structure brainstorming

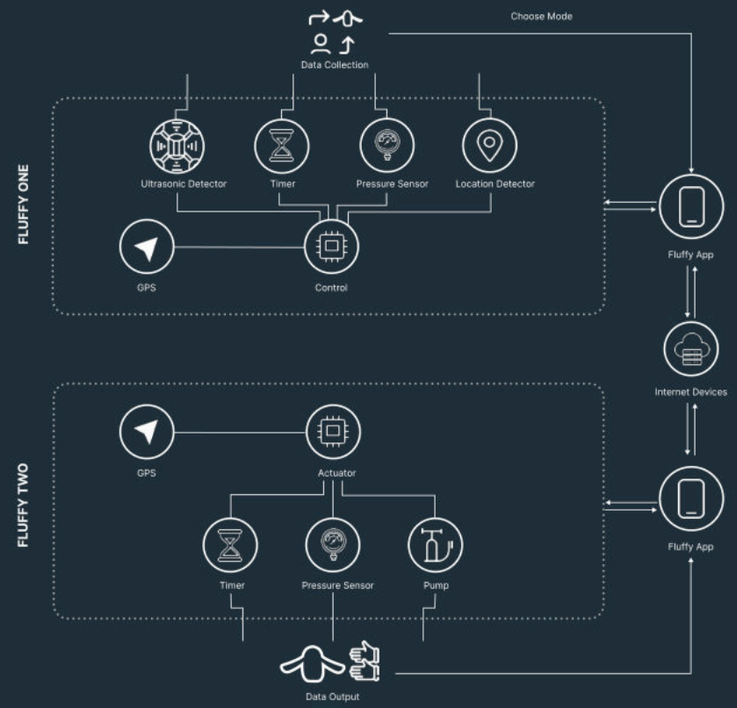
Connection

Cloud + Internet + Terminals is the mainstream of telecommunication, which means that hardware terminals such as phones and sensors, and software terminals such as apps are connected through the internet and cloud. I use this concept for reference. For people to hug each other over distances, people hug Fluffy, which transfers data to the user's app through Bluetooth. Apps exchange data through servers, thus, connecting users.



Hug Deposit & Realtime Hug

The Fluffy app provides an interface for users to control Fluffy and send data to users on the other end. Fluffy collects data so that the other Fluffy can execute the data, sending hugs.



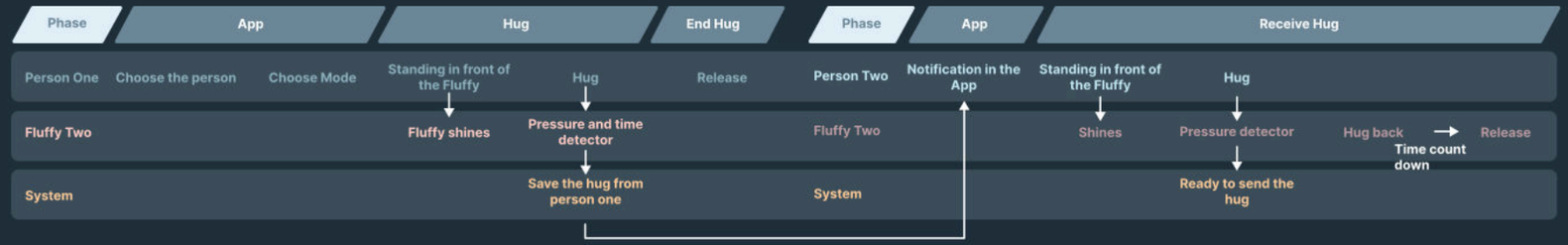
Interaction Design

Interaction Mode

Considering different scenarios, Fluffy has three modes: self hug, hug deposit, and real-time hug.

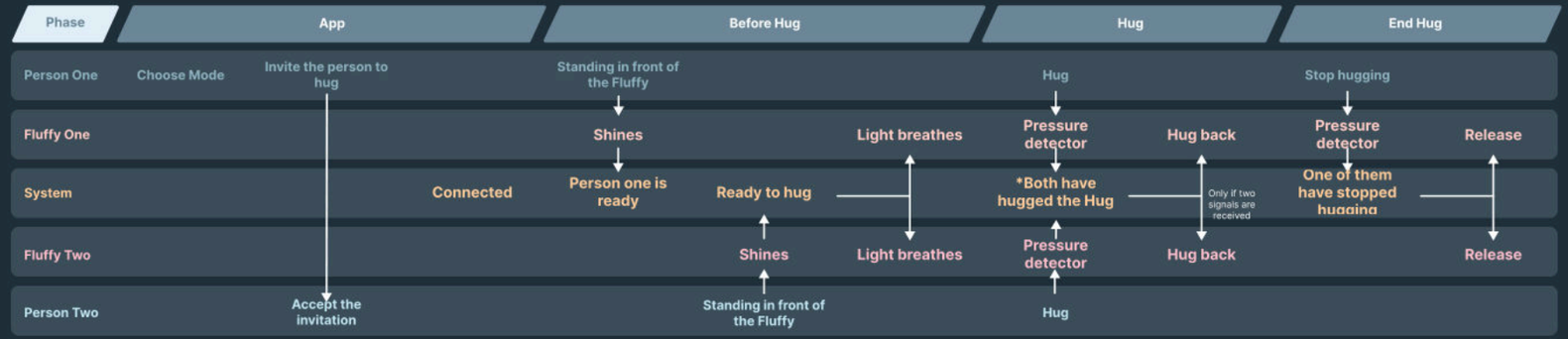
Mode 1: Hug Deposit

In the hug deposit mode, a person can send a hug to the other person like ordering a gift, in case that two people cannot hug in real-time. Fluffy will record the data from the sender and save it on the server. App will notify the receiver a hug is receievd. Once the receiver is ready to accept the hug, they can tap the App and perceive the hug from Fluffy at his end.



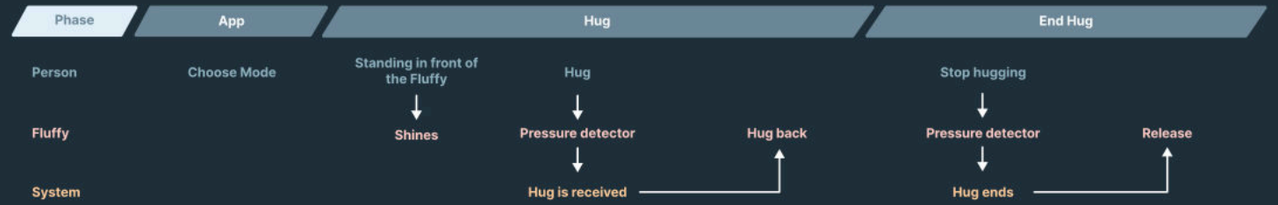
Mode 2: Real-Time Hug

In the real-time hug mode, people can hug each other over distances in the real-time. Hug received will be presented at the other end, creating an interactive, loving hugging experience



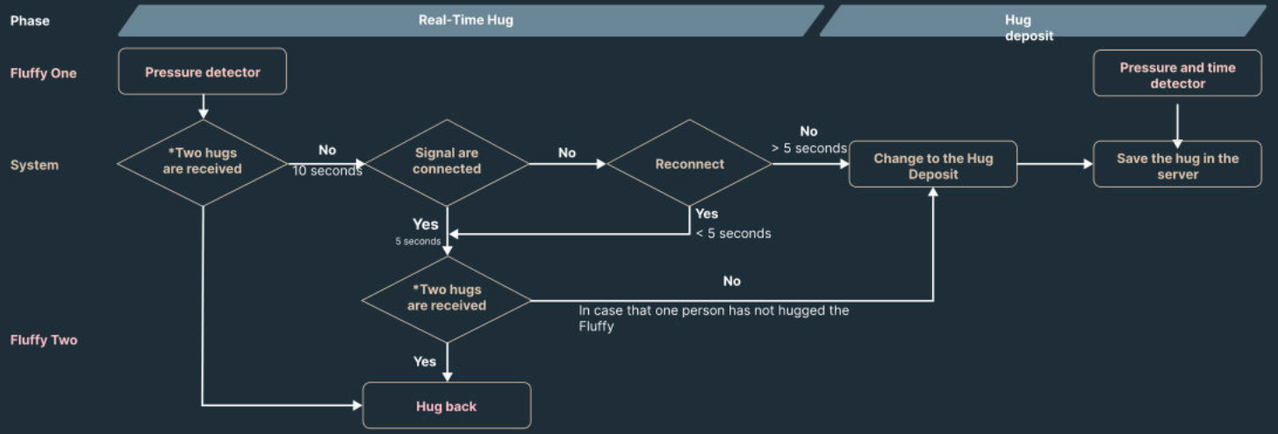
Mode 1: Self Hug

In the self-hug mode, people can give themselves a hug as a way of practicing self-compassion, as hugs increase feelings of love and kindness toward the self. As an important quality, self-compassion is positively correlated with an individual's overall psychological well-being. People who are self-compassion show less anxiety, depression, shame, and fear of failure. They experience greater social connectedness, emotional intelligence, happiness, and overall life satisfaction (Raypole, 2020).



Real-Time Hug Backup

Special occation: One of the users has not hugged the Fluffy or the signal has disconnected during the real-time hug mode due to unstable network.



App Design

♥ Design Principles

To provide a great user experience, I make the app's appearance and behavior integrate with its function.

Appearance: As a hugging app, it aims to give users welcoming and cozy feelings, so that I make the style cute, agreeable, and friendly, which resonates with the look of Fluffy. I also apply a warm color palette. I highlight content through the clear interface, precise icons, subtle adornments, and legible text. The design is light and airy, which aims to help users understand the content while not distracting their attention.

Behavior: To minimize the learning curve, I refer to the current messaging and calling app. I design virtual actions based on people's real-world experiences including drag and drop during categorization. To make users feel in control, I keep interactive elements familiar and predictable and allow users to cancel operations even though they are underway. I keep users in the loop by providing instant feedback for their actions. Touch enables access to functionality and content without losing focus.

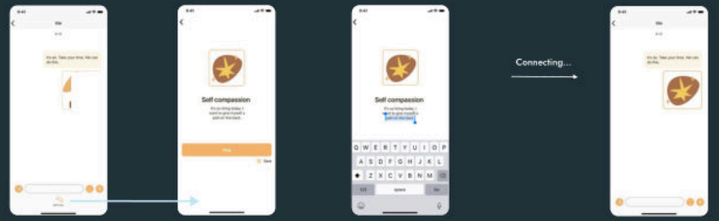
Function: The app aims to facilitate the connection between people, I emphasize its functions of communication and recording by applying more subtle, unobtrusive graphics, standard controls, and predictable behaviors.

♥ Interface Exhibition



Wireframe

Self Hug Mode



1. Choose the Dialog Box with Myself and Click Self Love

Sending a hug is as easy as sending a message. To hug oneself, the user can choose the dialog box with oneself, click the plus button, and start real-time hug by choosing the self love mode.

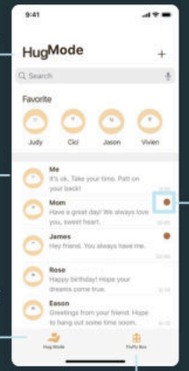
Before the hug, the user can choose whether they want to save the hug, edit the title and message, and upload a photo. The user can write down their mood or words of encouragement.

2. Feedback

Self compassion is done! A hug is sent.

Home Page

Users can edit and choose their favorite people for easy search.

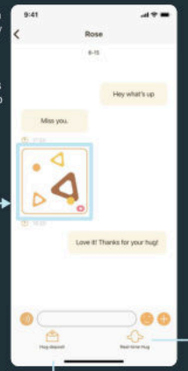


In Hug mode, users can choose to hug themselves, send hugs to loved ones, or hug someone in real-time

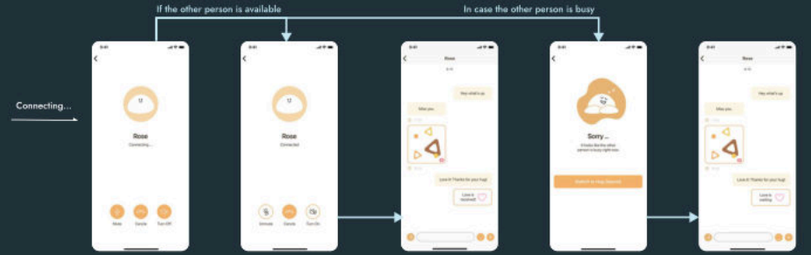
To receive the hug, the user can click the photo and get ready by standing in front of the Fluffy.

In case the user wants to save the hug, the user can long press the photo and wait for a heart to appear beside the photo. Then the hug is added into the Fluffy Box, where users can further edit and categorize hugs.

When a new message or hug is received, the app will notify users and mark the dialog box with a dot.



Real-Time Hug Mode



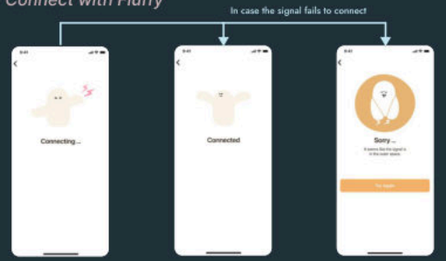
1. Send a Hug Invitation

Starting a real-time hug is the same as inviting someone for a voice chat. Both ends can communicate to get ready and talk emotionally when hug.

2. Interaction Feedback

In case the other person does not accept the invitation, the user has the option to switch to hug deposit mode and send a hug instead. Whether the real-time hug has successfully been received will leave a record in the chat box.

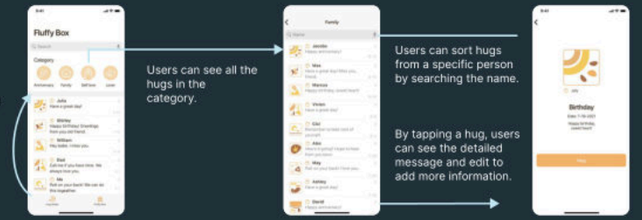
Connect with Fluffy



The app automatically connects with Fluffy when users opening the app. In case the connection fails, the app will reconnect during the hug process.

Browse and Receive a Hug in the Past

In Fluffy Box mode, users can browse hugs they have saved. They can categorize hugs and re-experience hugs they have received in the past.



Through drag and drop the hugs in the associated category, users can categorize hugs based on their customized tags.

Users can see all the hugs in the category.

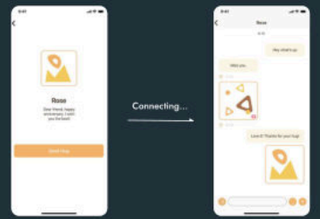
Users can sort hugs from a specific person by searching the name.

By tapping a hug, users can see the detailed message and edit to add more information.

Hug Deposit Mode

1. Edit the Message

Sending a hug is the same as preparing a gift for loved ones. Similar to Self Hug mode, Users can write down messages and upload images.



2. Love is Sent

The app with notify the receiver that a new hug is received.

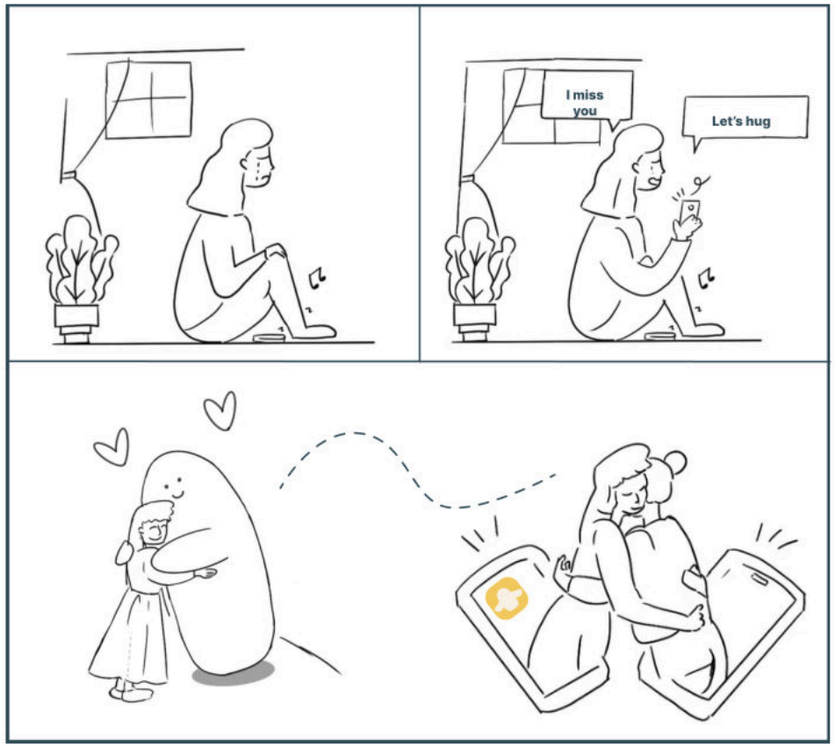
Scenarios

Users can either purchase Fluffy for home use or go to the public space and use Fluffy in the booth. Since users may not be at home when they really need some emotional comfort, they can find the nearest booth, connect with Fluffy, and get a moment of psychological comfort.

They can enter the booth through Near-field communication by tapping their phone to the door. A booth is a cozy place for people to take a break and connect with their loved ones.







Psychology Experiment

Since Fluffy is designed to connect people in distance, does Fluffy really get the job done? Thus, I design a psychology experiment to verify.

Procedure design

The scientific method, a problem-solving approach in sciences, has five basic steps and one Iteration step.

Make an observation	Hugging humans and intimate objects bring physical and mental benefits.
Ask a question	Can hugging love ones through intimate objects bring more health benefits?
Form a hypothesis	Understanding that hugs are from love ones make people happier.
Make a prediction	Receving love ones' hugs through Fluffy positively impacts people's mood.
Test the prediction	

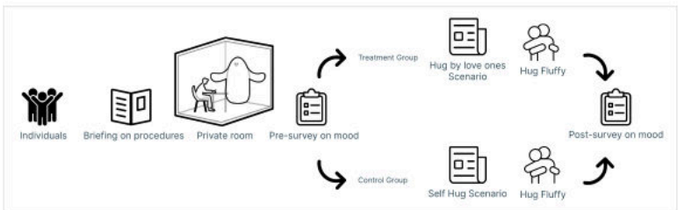


Since I want to test whether hugs from love ones have bigger health benefits, the IV is a hug's attached sender's information. As Fluffy is designed for home use, receiving hugs should be normal in daily life. Thus, I control the IV through scenarios to create an immersive use experience.



Based on comparison, I choose survey to measure health impacts. The survey will be structured with closed questions to improve consistency of responses.

I will randomly sample 20 individuals from school, with 4 students from five classes to represent my school. These 20 individuals will be randomly assigned into two groups, 10 individuals each.



Potential problems and solutions

Dishonesty

Since participants may guess experiment's objective through clues in an experiment and act to meet the objectives, I will make the surveys anonymous and ask them to put the finished surveys in the sealed box. Also, in the initial briefing, I will give them a false objective by stating that the experiment intends to test whether hugging inanimate object impacts people's mood.

Low response rate

To increase response rate, I will make the survey easy to complete by making the survey short and focused, right after the experiment. Question wording will be easy to understand. Also, I will set the right expectation about how long the experiment and surveys are going to take.

Missing Data

Since missing data distorts findings, I will eliminate missing data to preserve results. Since participants are few, I choose Pairwise Deletion, which only eliminates data when a vital data point is missing to save more data, compared to Listwise Deletion, which deletes the entire data.

Extraneous variables (EV)

EV refers to variables, other than the independent variable, that could affect an experiment's results. They should be controlled where possible. Thus, the experiment will be held in a private room to reduce interruptions.

Briefing on procedures

To reduce experimenter effects and ensure that researchers will not become a variable that accidentally influence the participant as well as test results, participants will receive the same briefing on the notes.

Thank you for participating in this experiment on hugging inanimate objects. The experiment will last 10 minutes. You will enter a private room, fill an anonymous survey, drop the survey in the box, and then read a scenario. You can then hug Fluffy, the installation in the room. The hug will last 20 seconds for the effect. And then you will fill another survey.

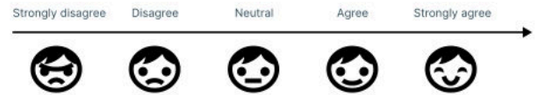
Survey Design

To measure mood, I refer to validated survey with a satisfying factorial validity and internal consistency. The survey is based on a widespread instrument, the Profile of Mood States (Morfeld, Petersen, Krüger-Bödeker, von Mackensen & Bullinger, 2007).

To ensure that targets complete surveys honestly without feeling burdensome, I make the survey short enough to cover the criterias: vigour, irritability, and fatigue.

How do you feel right now?	Vigour	Irritability	Fatigure
Lively	●		
Vigorous	●		
Engergetic	●		
Cheerful	●		
Alert	●		
Full of pep	●		
Active		●	
Nervous		●	
Angry		●	
Annoyed		●	
Peeved		●	
Spiteful		●	
Bad-tempered		●	
Furious			●
Lifeless			●
Weary			●
Exhausted			●
Sluggish			●
Worn out			●
Fatigire			●

I will use likert scale to evaluate the extent. The intensity will be reversed for some questions to ensure the validity of the data.



Scenario Design

Since the experiment intends to test whether receiving hugs from love ones through Fluffy has better impact, two groups of subjects will experience self hug or hug with their love ones. Both groups will given scenarios first, following instructions to hug Fluffy.

You feel so tired today after a day's work. You have dealt with some difficult clients. Although it took a lot of your engergy, you got it done nicely. After staying late to handle the extra work, you finally got home. You really need a pat on your back.

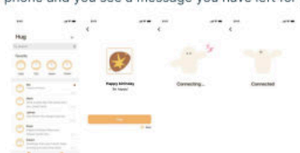
For treatment group

You take out your phone and you see a message from your mom.



For control group

You take out your phone and you see a message you have left for yourself.



You can receive the hug by standing in front of Fluffy and hug Fluffy.